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After Arion, Hermes, and a Free-Egress Argument: SN75 Hippius in May 2026

Eight months after Alphanomics: a new storage engine (Arion), a cross-subnet messenger (Hermes), a second region, and a repositioning from object store to AI data plane. What that means, and what is not yet settled.

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When this magazine last covered Subnet 75 in late September of 2025, the framing was Alphanomics, Hippius's incentive design and the way hAlpha routed real product revenue back to miners, validators, and stakers. The framing was correct for September. It is incomplete for May. In the eight months since, Hippius has replaced its storage engine, shipped a separate cross-subnet messaging protocol, opened a second geographic region, and quietly repositioned itself from a decentralized object store into something closer to a data plane for Bittensor's AI workloads. The repositioning is worth taking seriously and worth interrogating honestly. Both are the same exercise.

What Actually Shipped, in Order The technical work breaks cleanly since the original article.

The first piece is Arion. Through late 2025 and into the first quarter of 2026, Hippius removed IPFS from its storage layer and replaced it with an in-house engine built on CRUSH-based deterministic placement, the same approach

underpinning enterprise systems like Ceph, and Reed-Solomon erasure coding at a ten-plus-twenty shard ratio. Any ten of thirty shards reconstruct a file; the network can lose two-thirds of its nodes and the data remains intact. The team published the technical rationale on March 10 of this year in a post titled "Why We Built Arion." Shards are now distributed across what the project reports as more than 459 independent miners.

The Reed-Solomon scheme deserves accurate framing. A 10+20 layout means total storage overhead is three times the original file size, the same as classical triple replication, not less. Arion buys recovery flexibility and predictable retrieval, not lower per-byte cost. The architectural comparison is to Ceph and Google's ColossusFS, both of which adopted erasure coding for the same reasons years ago. Hippius is applying enterprise-grade primitives to a permissionless network. That is a real engineering claim. It is not, by itself, a cheaper-per-byte claim.

The second piece is the developer surface. On March 18, the team published a tutorial framing Hippius as a drop-in replacement for AWS S3: same SDK, same boto3 client, signature V4, path addressing, full support

for buckets, objects, multipart uploads, ACLs, policies, presigned URLs, and lifecycle rules. On March 24, a follow-up positioned the network as "agent-ready" with an llms.txt index at docs.hippius.com that allows AI agents (Claude, GPT, Gemini, or any open-source model with the same calling pattern) to consume the API surface without custom integration. The published examples cover LangChain document loaders, Claude tool-use with persistent memory, the OpenAI Agents SDK, and n8n. The pitch is unstated but obvious: every agentic AI workload accumulates working memory, scratchpad state, vector embeddings, retrieved documents, generated outputs. That memory has to live somewhere. The default is AWS. Hippius is making the case that for agents whose operators do not want their cognition tied to AWS account suspension, billing posture, or compliance policy, the default should be questioned.

The third piece is the multi-region rollout. On April 27, Hippius opened a second geographic region. Europe is now served from eu-central-1.hippius.com, set as the default that s3.hippius.com resolves to, and the United States from us-east-1.hippius.com. The pitch on X was three lines of boto3 code to migrate. The underlying point is that a decentralized storage network that wants to compete with hyperscalers has to behave like one at the developer interface, which means region selection that maps onto the data residency and latency assumptions an enterprise application is already making. The S3 surface plus regional endpoints is the team's bet that those assumptions translate, and that the friction of moving a workload off AWS should be measured in lines of code rather than weeks of integration.

The fourth piece is Hermes, currently in V0.2 early access. Hermes is not storage. It is a peer-to-peer data messenger with two transports: direct QUIC streaming for real-time gradient exchange between subnet miners, and an Arion Sync-Engine path for routing 100GB-plus tensor payloads through the storage layer. Encryption uses NaCl SealedBox (X25519 plus XSalsa20-Poly1305) applied in Rust before data hits the wire. Node identity is derived from Ed25519 keys mapped to SS58 addresses through an on-chain AccountProfile pallet, so Hermes peers are already addressable within the Bittensor account model. There is a Python SDK with a PyO3 backend that releases the GIL during heavy cryptographic work. The protocol is cross-subnet by design; the documentation's quickstart example shows a node on Subnet 42 talking to a node on Subnet 69.

The fifth piece is the new Sync Engine, which is migrating today, May 12, at block height 55,577,490, approximately 15:00 UTC. The team describes it as a rebuild from scratch to replace the previous version, which it characterized as slow and conflict-prone. As with all live migrations on a production network, the test is whether it ships cleanly and whether the existing data layer is preserved through it. By the time this article reaches readers in its second or third hour, that question will have an answer.

Why Hermes Is the More Interesting Bet The agent-ready framing is the one the team's marketing prefers. The Hermes bet is the one a careful reader should pay closer attention to.

In March, the Subnet 3 team, then operating as Templar, completed Covenant-72B, the largest decentralized large language model pre-training run ever performed. Seventy-two billion parameters, roughly 1.1 trillion tokens, more than seventy independent contributors connecting commodity GPUs over standard home internet. The model scored 67.1 on MMLU, putting it in the same performance band as Meta's LLaMA-2-70B, which was trained on a hyperscale cluster with infrastructure costs measured in the hundreds of millions. The technical innovation that made the run feasible was a compression protocol called SparseLoCo, which reduced inter-node communication by a factor of 146 through sparsification, two-bit quantization, and error feedback. SparseLoCo solved the algorithmic side of distributed training over the public internet. Hermes is built to solve the transport side.

That alignment is not coincidental. The bottleneck for any large permissionless training run is the same bottleneck Hermes was engineered for: encrypted, peer-to-peer, predictable-latency movement of model artifacts between independent miners who may sit on different subnets and different continents. Gradients during training, checkpoints between phases, dataset distribution at the start, weight delivery at the end. A trillion-parameter run, which renamed Teutonic team is targeting for later this month, multiplies every one of those payload sizes by an order of magnitude. A storage layer that cannot move data fast enough between participants becomes the new bottleneck even if the math works.

Whether Teutonic or any production training pipeline actually adopts Hermes is a question the public record has not yet answered, and it is the question that matters most for the thesis. The architectural fit is real. The adoption is not yet evidence. If a permissionless trillion-parameter training run completes in the next six weeks using infrastructure that includes Hermes in its data path, the story changes. If it completes using something else, the story is that Hippius built credible infrastructure for a workload that chose other rails. Both outcomes are possible. The reader should hold both.

The Egress Argument, Quantified Six days before this article, the Hippius account posted what it called a hot take but reads better as a thesis: "the AI

compute wars will be decided by storage." Model weights, training datasets, inference logs. All of it needs cheap, fast, reliable storage that does not call home to Amazon. Two days later, the team put a number behind the argument. One hundred terabytes of egress served across the Hippius network in the past month. AWS S3 would have billed approximately $8,000 in bandwidth for the same outbound traffic. Hippius billed zero. Egress is free, the team's framing being that "the cloud tax is optional."

The line worth pulling on here is structural, not promotional. Centralized cloud margins are not built on storage. They are built on outbound bandwidth. AWS, Google Cloud, and Azure all price ingress free and egress at rates that compound with workload scale, which is why migrating off the major clouds becomes structurally more expensive the more successful an application becomes. Cloudflare's R2 reshaped a piece of that market by zeroing egress and forcing the conversation back to actual storage cost. Hippius is making the same argument inside the Bittensor ecosystem, with one extra piece: because the cost structure is supposed to derive from a decentralized network of independent miners earning subnet emissions rather than from a single operator absorbing bandwidth as a loss leader, the pricing model is not the same as R2's even if the user-facing economics rhyme.

For AI workloads specifically, this matters more than it does for general object storage. A model-serving infrastructure delivers its weights to inference nodes on every cold start. A training pipeline moves checkpoints between participants. A retrieval-augmented generation system pulls documents on every query. All of that is outbound bandwidth. storage layer that does not meter it is structurally different from one that does, in ways that compound across workloads decentralized AI actually wants to run. One hundred terabytes a month is a real number. It would be a meaningful workload at any storage provider. But it is not yet the scale at which the question "do free-egress economics survive production model serving" gets answered. The honest read is that the structural argument is sound, the early data is consistent with the argument, and the test arrives at a workload order of magnitude or two above where SN75 currently sits. The team's framing should be evaluated when that test arrives, not before.

The other layer of honesty owed here is the same one this magazine has called for at the ecosystem level. The Pine Analytics work earlier this year put numbers on a structural problem the ecosystem had been gesturing at without quantifying. Network-wide, Bittensor miners receive on the order of $148 million in annual TAO emissions against what Pine could identify as $3 to $15 million in confirmed external revenue. Chutes (SN64), the most-cited subnet by usage, sits at roughly $52 million in annual emissions against $1.3 to $2.4 million in measured external revenue, a subsidy-to-revenue ratio between twenty-two and forty to one. Targon (SN4) is closer to 1.7 to 1, the best in network. Without those subsidies, Pine's modeling suggests Chutes's pricing would be between 1.6 and 3.5 times more expensive than DeepSeek or Together AI rather than cheaper. The "X times cheaper than AWS" framing that has become a recurring shorthand across the ecosystem, including SN75's own per-byte storage claims in our September article, is an emission-funded number until separated by audit.

Hippius's own publicly tracked PnL figures around $4.48 million put SN75 among the relatively healthier subnets on this metric, well clear of the worst of the subsidy ratios Pine documented. That is genuine and worth crediting. It does not, however, substitute for an audit-grade separation of usage-funded revenue from emission-funded subsidy on the storage layer specifically, the question of whether free-egress economics survive once the emission line is taken out of the cost stack. Until that separation is published, "free egress" should be read as a real and substantive product feature

and an unverified cost model at the same time.

The Bridge to the Closed Labs There is a bridge worth naming explicitly. OpenAI, Anthropic, Google DeepMind, and Meta have all converged on a stack where compute, storage, training orchestration, and identity sit inside a small number of vertically integrated providers. The artifacts produced by those labs (weights, datasets, evaluation logs, fine-tuning corpora) live in cloud storage owned by the same handful of companies that provide the compute. The integration is the product as

much as the model is.

The open-source alternative cannot replicate that integration accidentally. Someone has to build each layer, and the layers have to interoperate well enough for an end-to-end workload to function without anyone touching AWS. Bittensor's compute layer has Chutes and Lium. Its training layer, post-Covenant, runs on

Teutonic, Templar's community continuation, and Macrocosmos. Its inference layer has Targon and Ridges. The storage and data-transport layer is where SN75 is contesting. Whether it wins that contest depends less on what the team announces and more on whether the trillion-parameter run, when it happens, names Hippius in its infrastructure stack, or, more interestingly, whether it does not have to name anyone because the data plane has become invisible enough to assume.

Centralized labs have spent a decade making infrastructure assumptions invisible. The open-source side is at the stage where every layer still requires explicit selection. Hermes plus the multi-region S3 surface is a bid to start collapsing that selection at the data plane.

The Covenant Context, Read Through AI Any honest article on a Bittensor subnet in 2026 has to engage with what happened on April 9 and 10. The crypto-publication framing of that event focused on price action. The AI-publication framing has to focus on underlying question: can a permissionless decentralized network actually serve as durable infrastructure for researchers and builders, or are the operators of any successful subnet contingent on the discretion of a small group at the protocol's center?

Sam Dare's public exit from Bittensor, walking out the team that had just shipped Covenant-72B, the network's single most credible AI achievement to date, answered that question in one direction. His allegations against Jacob Steeves alleged unilateral suspension of emissions, override of community moderation, and "decentralization theater." Steeves disputed the specifics. What is not disputed is that the operator who produced the most important model on network felt unable to continue building on it. That should concern anyone treating Bittensor as serious AI infrastructure. The protocol's response was BIT-0011, the Conviction Mechanism rolling out tomorrow, which

makes subnet ownership a function of locked-alpha conviction scores over 30-day intervals. By Steeves's own account, the proposal was drafted by Dare before his exit.

The community then restarted SN3, SN39, and SN81 from open-source code without operator involvement and renamed the Templar work Teutonic. That answered the question in the other direction. The AI work resumed. The training pipeline did not depend on any single team to continue. Antifragility is a strong word and the test of whether it actually applies is whether Teutonic delivers the trillion-parameter run on the timeline the community has committed to.

The relevance for Hippius is twofold. First, SN75's miner-lockup architecture under Alphanomics already operates on principles structurally similar to what BIT-0011 imposes network-wide. Miners forfeit dividends on locked hAlpha as collateral. Hippius was early to the pattern the protocol is now formalizing. Second, SN75 runs its own Substrate chain alongside its subnet, which creates a different governance surface than subnets that exist purely as Bittensor metagraph entries. Whether that structural difference insulates SN75 from operator-level capture, or merely shifts the locus of that question to the Hippius validator set, is something the public record cannot yet answer. Hippius has not been stress-tested on that axis. Most subnets have not.

The native WASM bridge contract on the main Bittensor chain, pending Opentensor approval in our September article, has not shipped in the eight months since. The provisional bridge runs, the alignment

between Bittensor-side Alpha and Hippius-chain hAlpha continues to depend on off-chain machinery, and a keystone protocol upgrade has been silent for two-thirds of a year. For an AI builder evaluating whether to commit to SN75 as a storage backend, that silence is information.

What to Watch The honest read on where SN75 sits. The team has shipped real product across multiple surfaces in eight months. Arion is a substantive rewrite of the storage layer using primitives borrowed from enterprise systems. Hermes

V0.2 is a coherent attempt to capture the cross-subnet data exchange use case that any large-scale decentralized training run will need. The multi-region S3 surface is a credible developer strategy. The free-egress argument has a real number behind it now and a sound structural case underneath it.

What is not settled. Hermes is V0.2 early access with no documented production adoption by other subnets yet. Arion's storage format is, by the team's own GitHub README, in active development with breaking changes possible. The Sync Engine migration ships today; clean execution matters and is not yet evidence. The WASM bridge is unresolved after eight months of silence. Performance benchmarks against S3, R2, Filecoin, and Arweave under representative workloads have not been published. The emission-versus-revenue separation that Pine Analytics put on the ecosystem agenda has not been audit-resolved for SN75 specifically. The post-Covenant governance environment has not been stress-tested at the operator level for any subnet that runs its own chain.

The questions that will actually determine whether SN75 is what it claims to be are knowable and testable. Clean execution of today's Sync Engine migration. Published benchmarks against centralized and decentralized storage incumbents. Hermes adoption by Teutonic or any other subnet running production training pipelines, ideally documented in the infrastructure section of a model release. The status of the Opentensor WASM approval. Whether revenue figures denominated in actual customer spend close the gap on emission-funded pricing. Whether free-egress economics survive a workload pattern resembling production model serving at scale. Whether the Hippius chain demonstrates operator-level resilience if it is ever stress-tested. Those are the data points this magazine will watching.

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